

Meta Quest 3s for dVRK Teleoperation

Introduction to Augmented Reality (EN.601.654/454)

Fall 2025

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Introduction

- The da Vinci Research Kit (dVRK) is a teleoperated surgical robot platform for research purposes based on the da Vinci Surgical System from Intuitive Surgical. Surgeons operate the Patient Side Manipulator (PSM) using the Master Tool Manipulator (MTM).
- The aim of this project is to implement an improved XR system utilizing the Meta Quest 3S and handheld controllers. With the controllers, two PSM arms can be manipulated as well as the endoscopic camera manipulator (ECM).

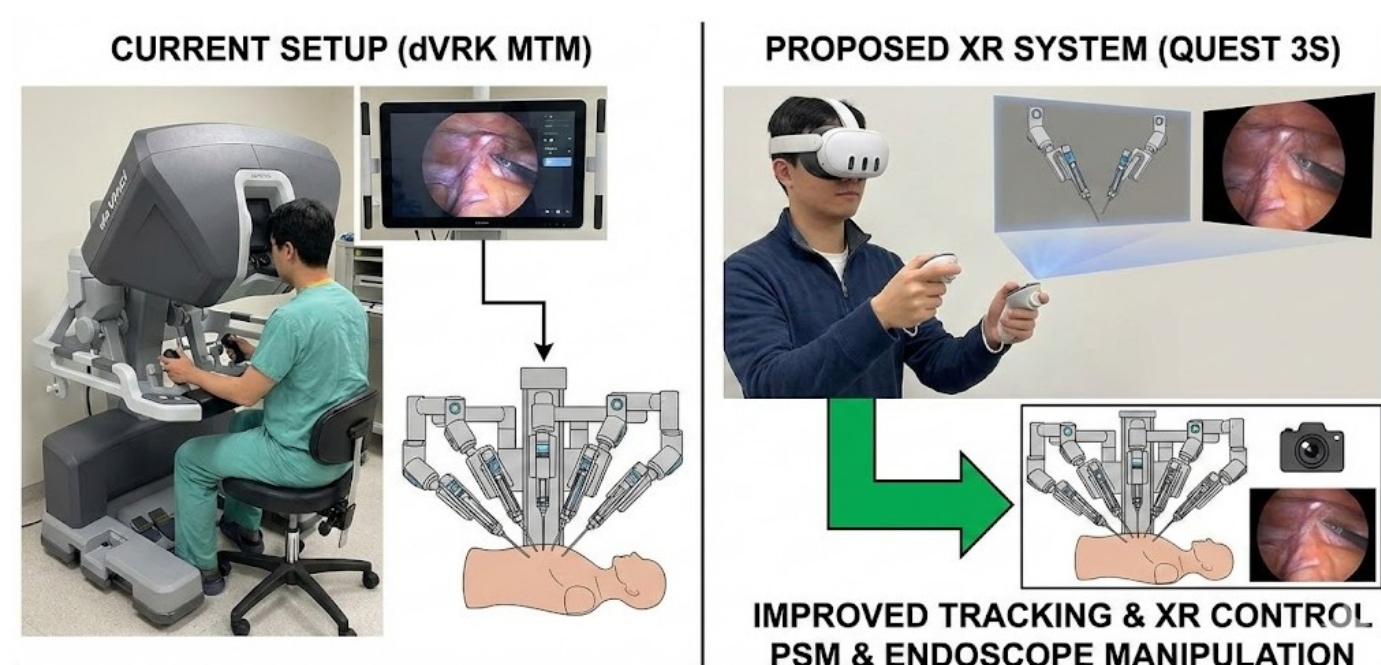


Fig. 1 dVRK MTM system vs. proposed Meta Quest 3s

The Problem

- Integration of XR control of the dVRK system for robot-assisted surgery can simplify complex procedures, while eliminating the need for the MTM at all -- increasing accessibility of the system by reducing cost and footprint.
- In the past, XR control of the PSM has been achieved with HoloLens 2 and hand tracking [1]. However, it is difficult to have accurate manipulation while including different tasks such as clutching and gripping.

The Solution

- An AR MTM environment built in Unity with video streaming and Quest 3s controller tracking for PSM/ECM control using a state machine (Fig. 2). The project is built on the PC rather than the headset itself to reduce latency.
- The endoscope video is livestreamed using the GStreamer library and using a Unity plugin, broadcast onto the Quest 3s display. Visual and text indicators above the video track the state of each controller individually.
- Hand controllers are visible below the endoscope stream for initial alignment. Users can switch between endoscope control and PSM teleoperation/clutch using the controller buttons (Fig. 3).

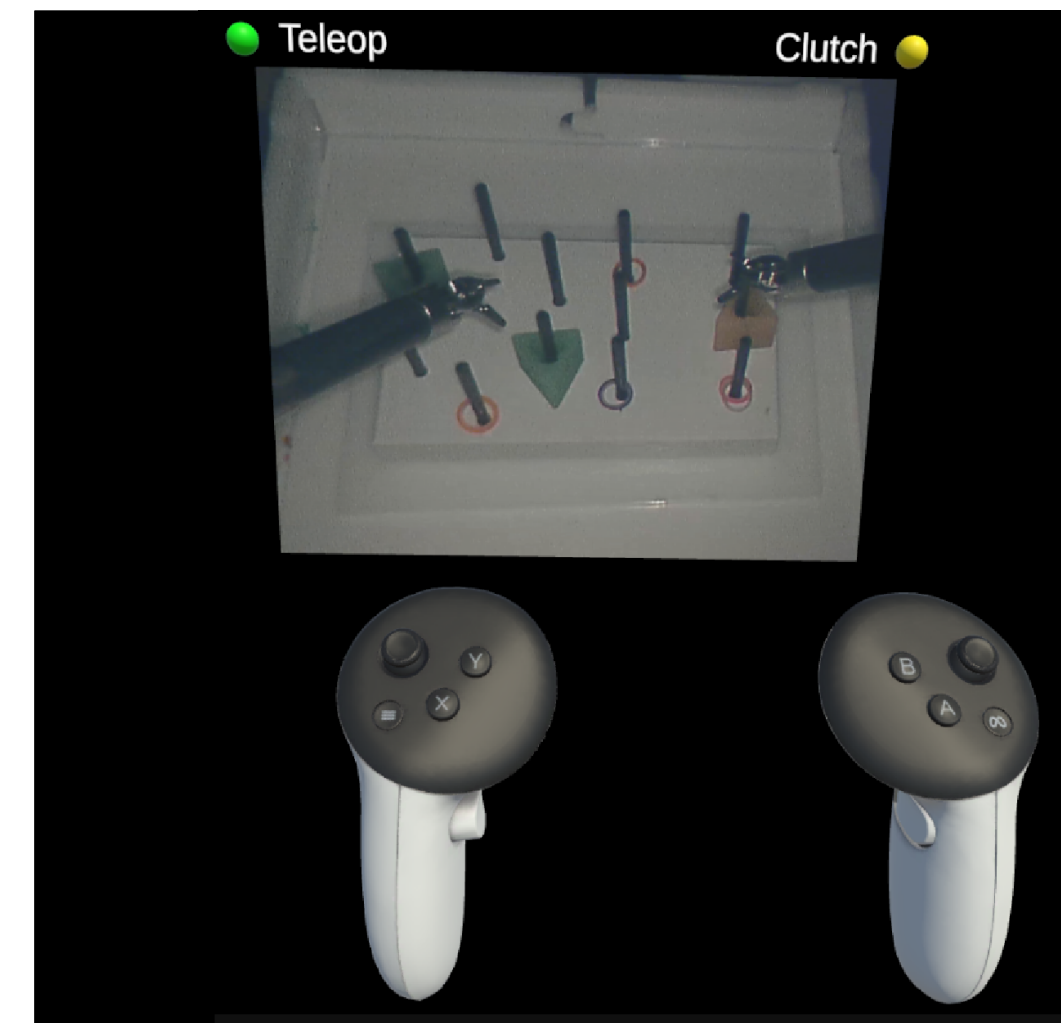


Fig. 4 User display with endoscope stream, controllers, and state indicators

Outcomes and Results

- We successfully enabled teleoperation of two PSMs and the ECM of the dVRK in an AR environment, eliminating the need for the dVRK MTM to control the system.
- Latency was tested by comparing timestamps through the ECM to Unity. We measured an average latency of 0.30 s. The latency from the Unity project to the Quest Link is reported to be 0.06 - 0.07 s [2], resulting in a total latency of approximately 0.37 s.

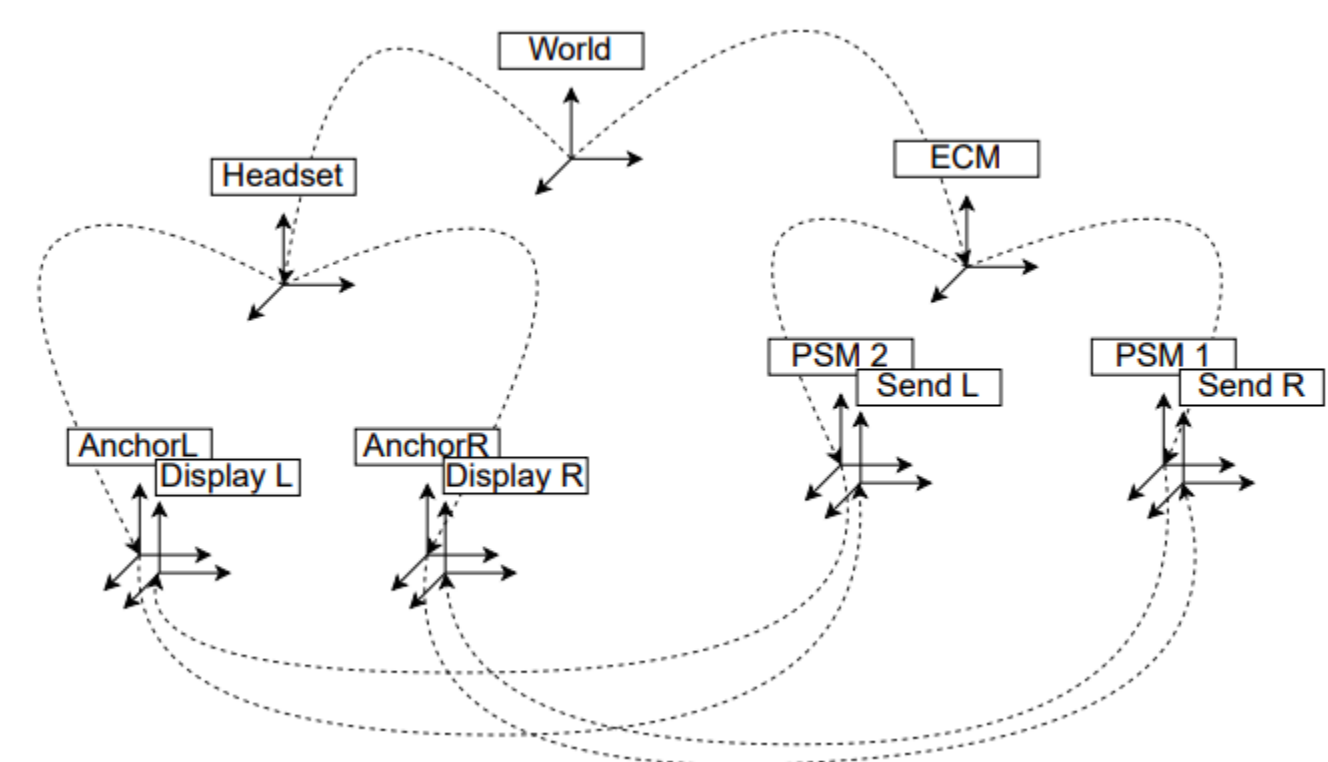


Fig. 5 Frame Transformation Setup for the AR MTM System

Future Work

- Reduce video streaming latency through improved compression methods.
- Enable improved absolute orientation tracking through hand-eye calibration.
- Enable stereo vision for improved depth perception.

Challenges & Limitations

- Current communication protocol for video streaming has limited bandwidth, resulting in noticeable visual latency.
- User is required to define the current orientation of each controller individually.

Credits

- Joseph: State machine design and implementation, controller clutching visualization
- Michelle: Endoscope video streaming integration, hand controller management, UI design
- Yub: PSM and ECM communication between Unity and dVRK as well as correction of transformation frames

Publications

- [1] M. Magnani, L. Cruciani, E. De Momi, and P. Kazanzides, "Console-Free Mixed Reality Teleoperation of the da Vinci Research Kit," ISMR, Apr. 2026 [under review]
- [2] H. Zhou, "Technical Comparison: DisplayPort Direct Connection vs. Quest 3 Streaming Solutions for PCVR," Pimax Store, Jul. 16, 2025. <https://pimax.com/blogs/highlights/technical-comparison-displayport-direct-connection-vs-quest-3-streaming-solutions-for-pcvr> (accessed Dec. 12, 2025).

Acknowledgements

- Thank you to Matteo Magnani, Anton Deuget, Nick Greene, and Haochen Wei for their guidance and support throughout this project.

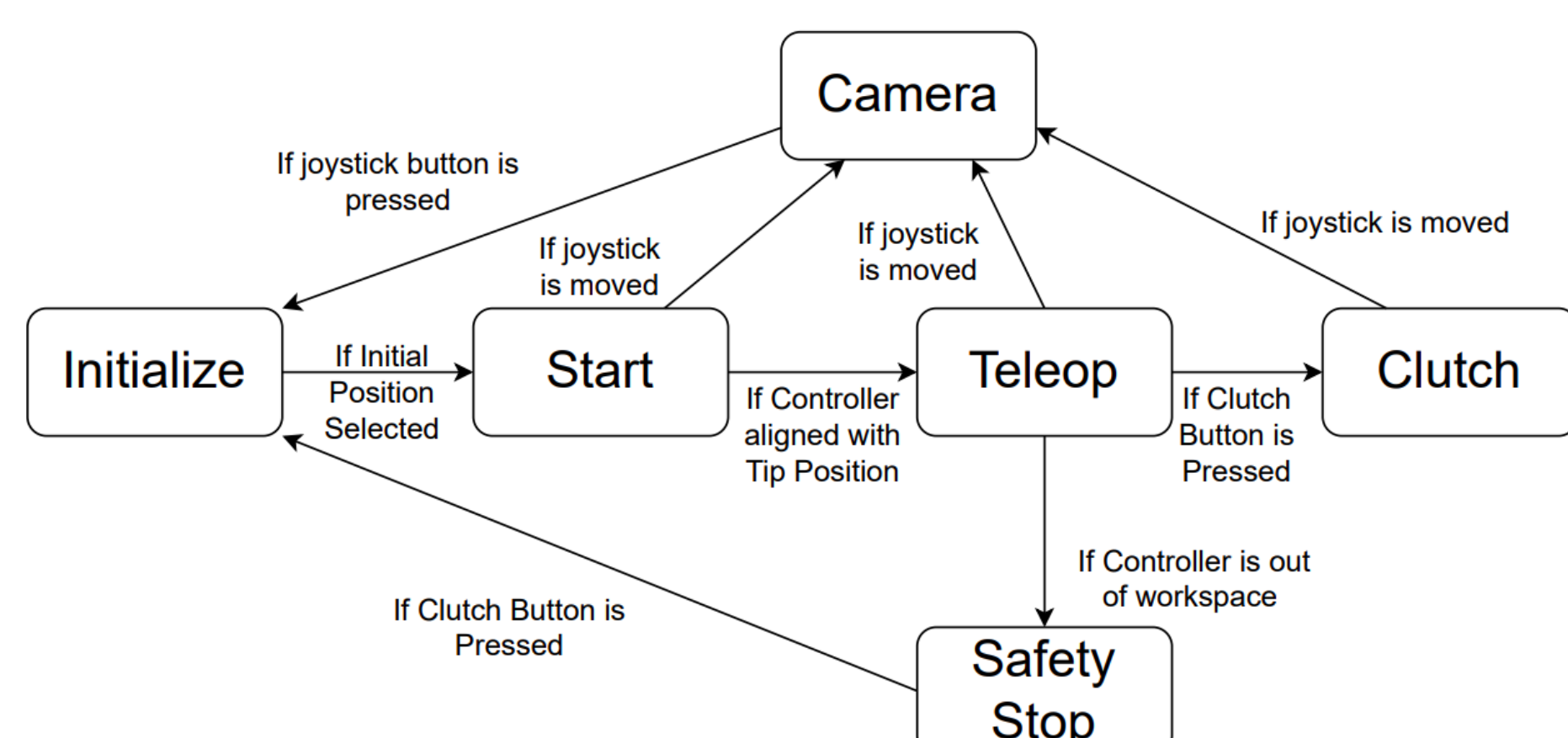


Fig. 2 State machine for dVRK Quest3s teleoperation

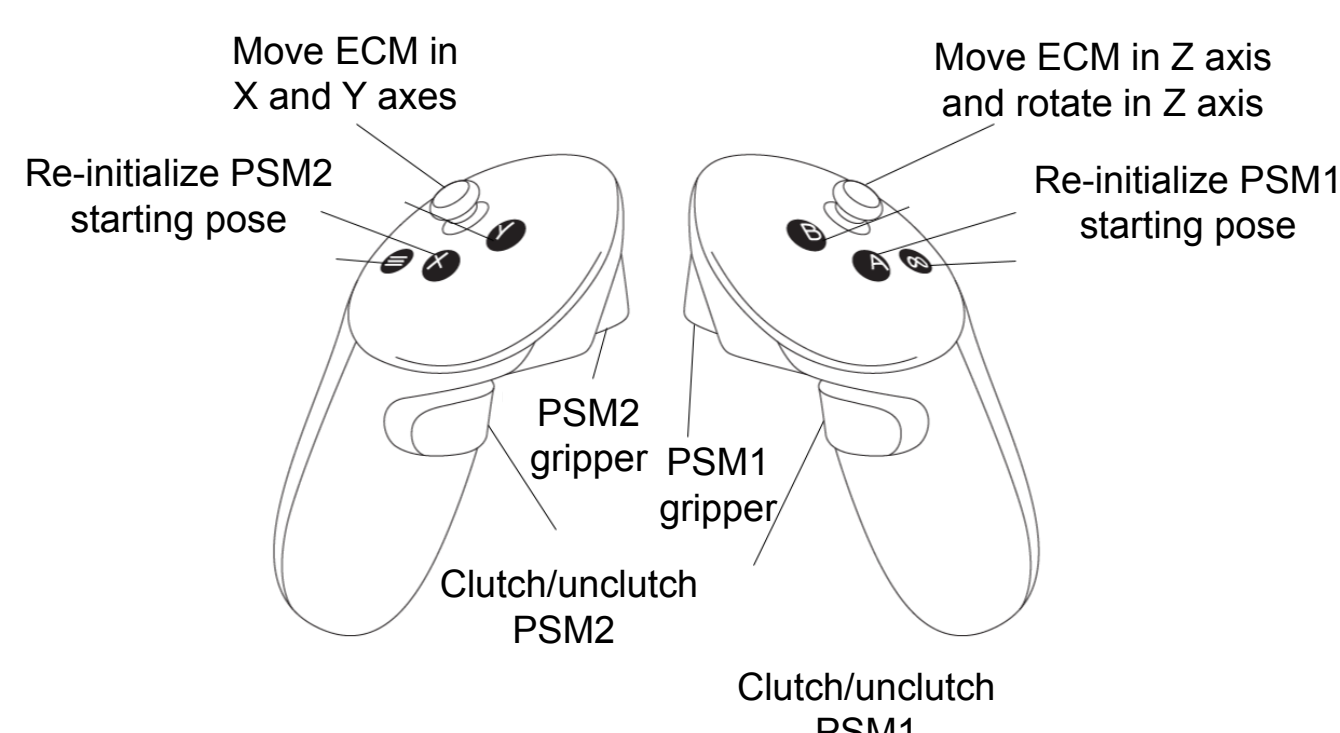


Fig. 3 Hand controller button operations